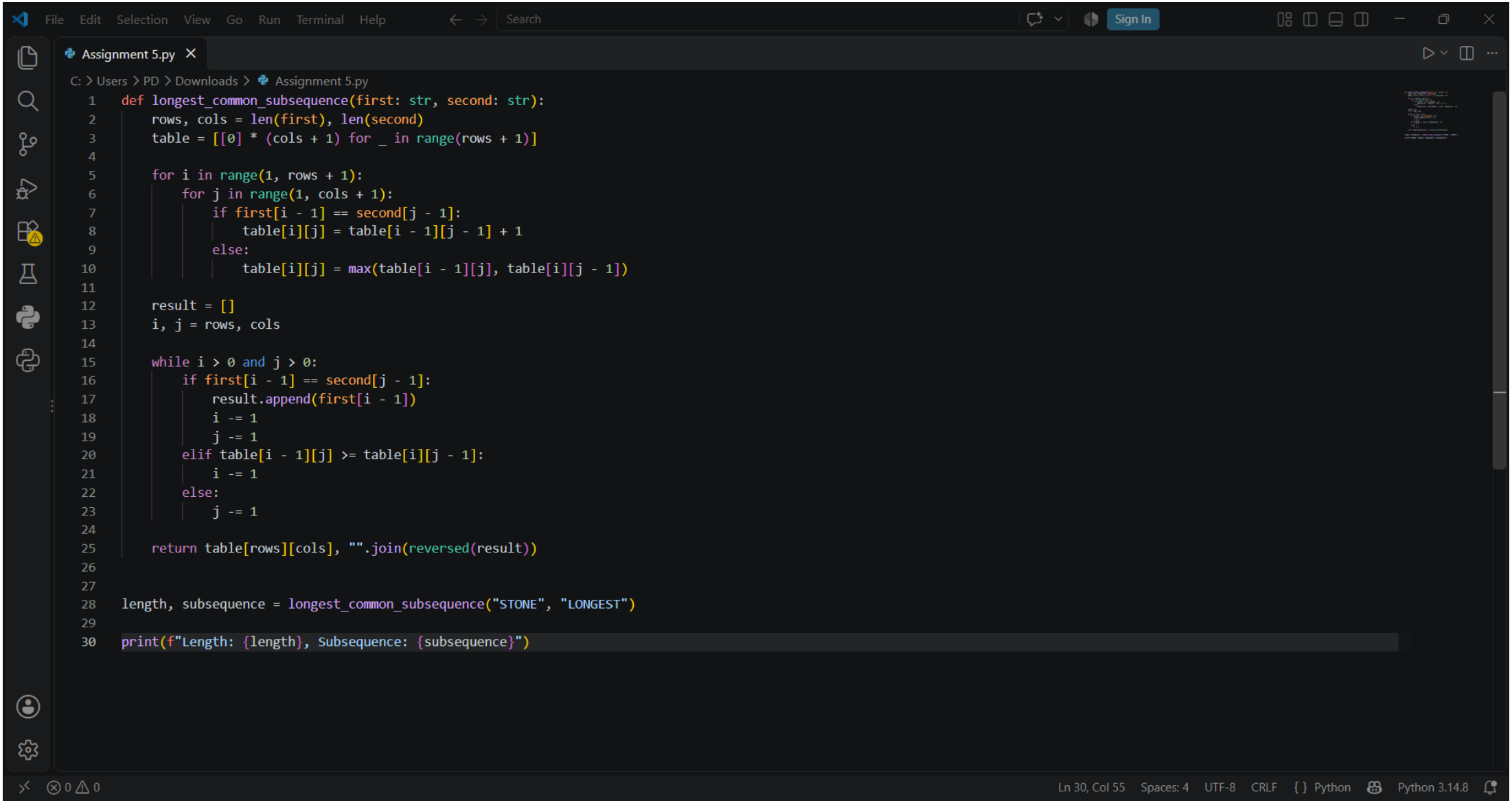
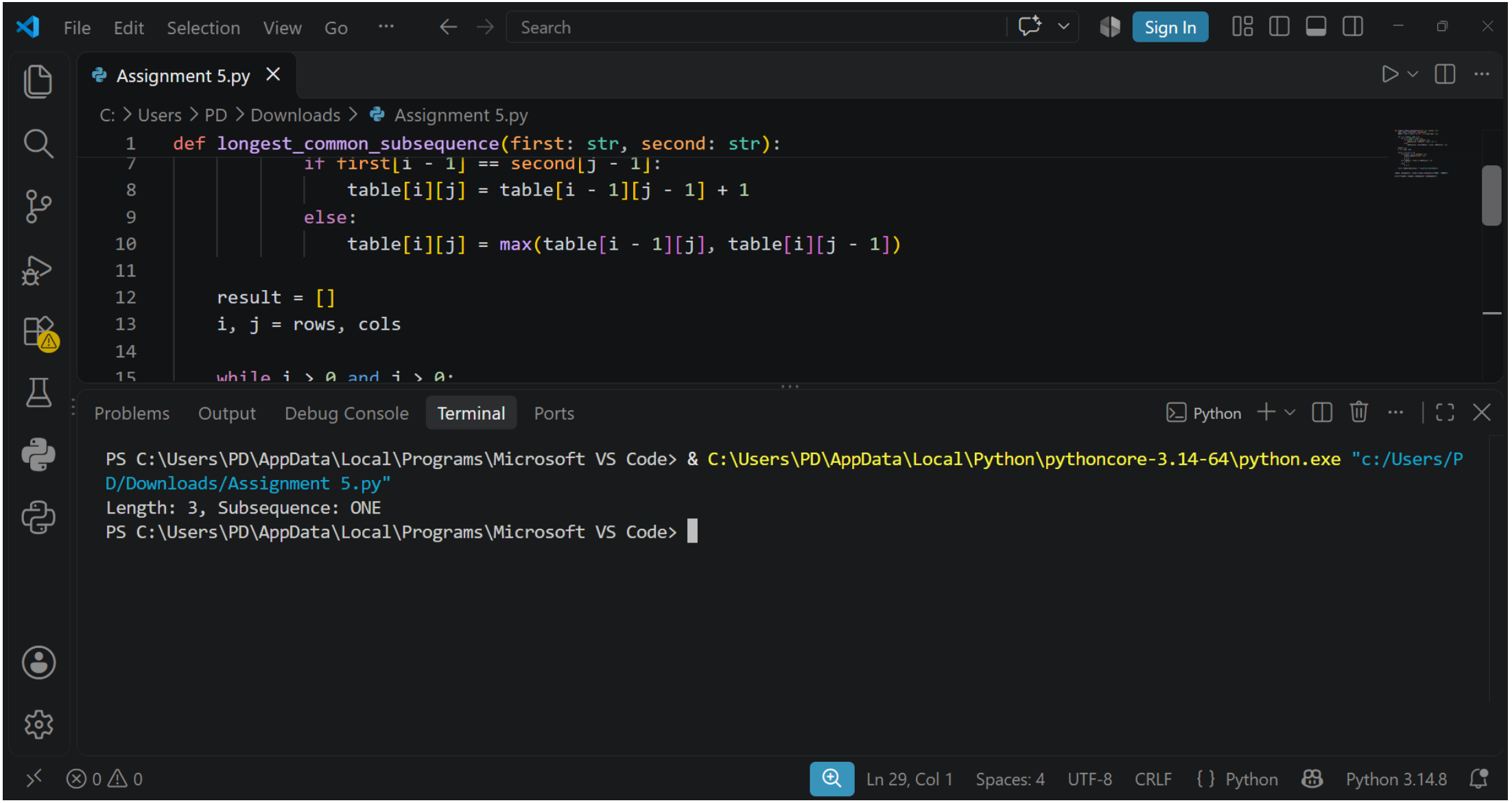

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Assignment 5.py
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def longest_common_subsequence(first: str, second: str):
rows, cols = len(first), len(second)
table = [[0] * (cols + 1) for _ in range(rows + 1)]
for i in range(1, rows + 1):
for j in range(1, cols + 1):
if first[i - 1] == second[j - 1]:
table[i][j] = table[i - 1][j - 1] + 1
else:
table[i][j] = max(table[i - 1][j], table[i][j - 1])
result = []
i, j = rows, cols
while i > 0 and j > 0:
if first[i - 1] == second[j - 1]:
result.append(first[i - 1])
i -= 1
j -= 1
elif table[i - 1][j] >= table[i][j - 1]:
i -= 1
else:
j -= 1
return table[rows][cols], "".join(reversed(result))
length, subsequence = longest_common_subsequence("STONE", "LONGEST")
print(f"Length: {length}, Subsequence: {subsequence}")
0 0
Ln 30, Col 55 Spaces: 4 UTF-8 CRLF { } Python Python 3.14.8


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Assignment 5.py
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def longest_common_subsequence(first: str, second: str):
if first[i - 1] == second[j - 1]:
table[i][j] = table[i - 1][j - 1] + 1
else:
table[i][j] = max(table[i - 1][j], table[i][j - 1])
result = []
i, j = rows, cols
while i > 0 and j > 0:
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Python
PS C:\Users\PD\AppData\Local\Programs\Microsoft VS Code> & C:\Users\PD\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/PD/Downloads/Assignment 5.py"
Length: 3, Subsequence: ONE
PS C:\Users\PD\AppData\Local\Programs\Microsoft VS Code>
Ln 29, Col 1
Spaces: 4
UTF-8
CRLF
Python
Python 3.14.8